



Transient thermal performance of multilayer thermal protection systems doped with phase change materials

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ABSTRACT

In this article, the transient heat transfer process of the multilayer thermal protection system (TPS) doped with the phase change materials (PCM) is studied. The multilayer TPS consists of the C/SiC composite material, alumina aerogel, silica aerogel, and silica aerogel doped with the PCM. A thermal lattice Boltzmann model is adopted to consider the effect of interlayer thermal contact resistance on the transient thermal response of the TPS. An enthalpy-based lattice Boltzmann model is adopted to solve the solid–liquid phase change problem. The transient heat transfer characteristics of two kinds of TPSs in different layouts are analyzed in detail. The results show that whether the doped PCM enhances the thermal insulation performance of the TPS depends on the layout design of the multiple layers and service time. There exists an optimal layer thickness of the PCM that makes a TPS have the best thermal insulation performance, and the optimal value increases with the service time. The interlayer thermal contact resistance can have a significant effect on the transient thermal response of the TPS, and the optimal value of the PCM thickness will reduce.

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1. Introduction

The thermal protection system (TPS) is designed to protect aerospace crafts from the ultra-high temperature environment due to the aerodynamic heating [1]. A typical TPS consists of a surface material with the high temperature capability and underlying insulation materials to inhibit the heat conduction. In the past decades, several materials were investigated to improve the performance of TPS at the ultra-high temperature, of which the carbon fiber reinforced silicon carbide (C/SiC) is considered to be the potential material to endure high temperature due to its high mechanical strength and low specific weight [2, 3]. As for the thermal insulation materials, nanoporous super insulation materials, such as the silica (SiO₂) aerogel and alumina (Al) aerogel, are widely adopted to suppress heat conduction because of their extremely low thermal conductivity [4–6]. Although the thermal insulation performance of Al aerogels is not as good as SiO₂ aerogels, Al aerogels have a higher mechanical strength and an enhanced chemical stability at the high temperature.

Given the advantages and disadvantages of aforementioned materials, it is beneficial to combine them together to form a multilayer TPS which can endure the ultra-high temperature and meanwhile have a better thermal insulation performance. The overall thermal insulation performance of the TPS depends on the effective thermal properties of aforementioned materials

(i.e., C/SiC composite materials, SiO₂ aerogels, Al aerogels) and the inevitable interfacial thermal contact resistance (TCR) between different layers. Hence, many efforts have been paid to predict the effective thermal conductivities of C/SiC composite materials [7–9], SiO₂ aerogels [10–12], and Al aerogels [13], as well as the interfacial TCR [14–16].

However, only a limited number of studies focus on the overall thermal insulation performance of the TPS. In this regard, Swann *et al.* [17] first proposed a one-dimensional finite difference method to analyze the transient thermal response of the TPS. Li and Cheng [18] predicted the effective thermal conductivity and steady-state heat flux of the multilayer perforated insulation material. Daryabeigi [19] developed a numerical model to investigate the optimum design of multilayer fibrous insulations. Shih *et al.* [20] used the finite volume method to simulate the transient ablation phenomena of multilayer insulation materials. Ji *et al.* [21] investigated the effect of the layout and the density of insulation materials on the thermal insulation performance of multilayer fibrous insulations.

The phase change materials (PCM) [22, 23] are the promising materials for the applications in the TPS since they can absorb a large amount of energy during the phase transition process [24, 25]. However, the thermal conductivity of PCM is generally higher than that of nanoporous aerogels. Thus, whether the PCM can improve the TPS' insulation performance still needs to be further discussed. Nevertheless, only few studies report on the transient thermal response of the multilayer insulation systems doped with the PCM [26], and they did not consider the effects of interlayer TCR on the overall thermal insulation performance. In this work, we develop a numerical model to analyze the transient thermal response of the multilayer TPS doped with the PCM with consideration of effects of interlayer TCR.

In the past decades, lattice Boltzmann method (LBM) has emerged as an efficient tool to solve the Navier–Stokes equations and diffusion equations [27, 28], and nowadays it has also been developed to simulate conjugate heat transfer in the heterogeneous media [29–31] and solve the solid–liquid phase change problem [32, 33]. In this work, the LBM is adopted to solve the transient heat transfer process involving the solid–liquid phase change problem in the multilayer TPS considering the effect of TCR. To the best of authors' knowledge, it is the first time to consider the effect of the TCR on the transient thermal response of the TPS.

This article is organized as follows. The descriptions and effective thermal properties of the TPS are introduced in Section 2. The numerical models to simulate the heat conduction with general boundaries and solid–liquid phase change problems are presented in Section 3. Validations are presented in Section 4 to show the accuracy of the present numerical model. Finally, some conclusions are drawn in Section 5.

2. Description of TPSs

In this work, the PCM is doped into the porous SiO₂ aerogel and two layouts of the multilayer TPS, as shown in Figure 1, are studied. For the layout A, the C/SiC composite material is put at the top side to endure the high temperature, and next by the Al aerogel, SiO₂ aerogel, while the layer of SiO₂ aerogel doped with the PCM at the bottom. For the layout B, the layers of C/SiC composite material, Al aerogel, SiO₂ aerogel doped with the PCM and SiO₂ aerogel are put in sequence from top to bottom.

The doped PCM is chosen to be the erythritol whose phase transient temperature is 117 °C. For the thermal properties of the erythritol, one can refer to Ref. [26]. The SiO₂ aerogel doped with the PCM is treated to be a homogeneous material with the effective thermal properties calculated by:

$$\rho_{\text{eff}} = 1 / (f_{\text{m},s} / \rho_s + f_{\text{m},\text{PCM}} / \rho_{\text{PCM}}) \quad (1)$$

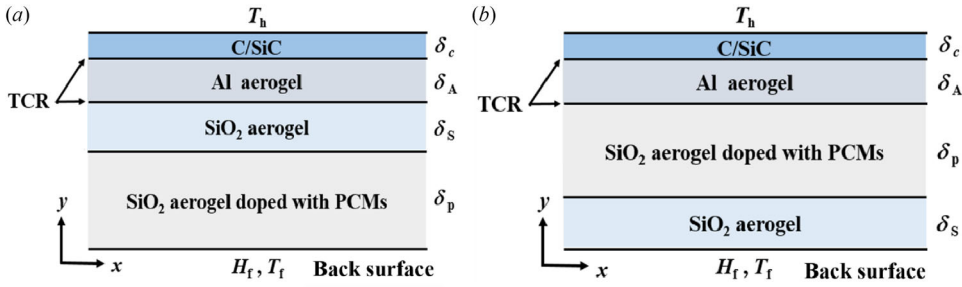


Figure 1. Two typical layouts of thermal protection systems: (a) Layout A and (b) Layout B. The thicknesses of the layer of C/SiC, Al aerogel, SiO₂ aerogel, and SiO₂ aerogel doped with PCMs are denoted as δ_c , δ_A , δ_s , and δ_p , respectively.

$$C_{p, \text{eff}} = \frac{f_{m,s} C_{p,s} / \rho_s + f_{m, \text{PCM}} C_{p, \text{PCM}} / \rho_{\text{PCM}}}{f_{m,s} / \rho_s + f_{m, \text{PCM}} / \rho_{\text{PCM}}} \quad (2)$$

$$L_{f, \text{eff}} = f_{m, \text{PCM}} L_{\text{PCM}} \quad (3)$$

where ρ , C_p , and L_f are density, specific capacity, and latent heat, respectively; f_m represents the mass fraction and is set to be 50% in this work.

The temperature-dependent thermal conductivities of SiO₂ aerogels and Al aerogel can be obtained by:

$$\lambda = 0.03987 - 6.29356 \times 10^{-5} T + 1.02587 \times 10^{-7} T^2 - 2.80066 \times 10^{-11} T^3, \quad (4)$$

$$\lambda = 0.00961 + 5.78739 \times 10^{-5} T - 8.56049 \times 10^{-8} T^2 + 4.87845 \times 10^{-11} T^3, \quad (5)$$

respectively, which are fitted from the experimental results presented in Ref. [26].

In this work, the overall thickness of multilayer TPSs is fixed at 0.05 m, with the thicknesses of C/SiC and Al aerogel given to be 0.005 m, 0.01 m, respectively. Thus, the total thickness of layers of the SiO₂ aerogel and that doped with PCM is 0.035 m. To find out the optimal thickness of the layer that doped with PCM, its thickness is varied from 0 to 0.035 m with a step of 0.005 m. The simulation parameters and other thermal properties of TPS are summarized in Table 1.

3. Numerical model

In this work, we assume that the extinction coefficient of the entire TPS is large enough that the radiative heat transfer inside the TPS can be neglected. Besides, the solid–liquid phase change is dominated by the conduction and the effect of convection is neglected due to the confine of nanopores. When subjected to the high temperature, the transient thermal response of the multilayer TPS includes the conductive heat transfer through the multicomponent media and the related solid–liquid phase change process. If the TCR between different layers is negligible, the temperature and heat flux should be continuous at the interface. Otherwise, a temperature difference induced by the TCR should be considered at the interface.

3.1. LBM for conjugate heat transfer equation with general boundaries

The governing equation for the heat transfer process is

$$\frac{\partial T}{\partial t} = \frac{1}{\rho c_p} \nabla \cdot (\lambda \nabla T) + S \quad (6)$$

where T is the temperature, ρc_p is the volumetric specific capacity, λ is the thermal conductivity, S is the source term. When the domain is subjected to a Robin boundary condition

Table 1. Simulation parameters and thermal properties of TPS [26].

	C/SiC	Al aerogel	SiO ₂ aerogel	SiO ₂ aerogel doped with PCM
Thickness	0.005 m	0.01 m	0–0.035 m	0.035–0 m
Thermal conductivity	1.5 W/(m·K)	Eq. (4)	Eq. (5)	0.059 W/(m·K)
Density	1500 kg/m ³	450 kg/m ³	350 kg/m ³	Eq. (1)
Specific capacity	800 J/(kg·K)	950 J/(kg·K)	1000 J/(kg·K)	Eq. (2)

$$\alpha_1 T + \alpha_2 \frac{\partial T}{\partial \mathbf{n}} = \alpha_3 \quad (7)$$

where α_i are the parameters which can be the functions of time and space, and $\mathbf{n}$ is the unit normal vector outward to domain.

A D2Q5LB model is adopted to solve Eq. (6), and the evolution equation of the temperature distribution function is

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)] + \Delta t \omega_i S \quad (8)$$

where $\mathbf{x}$ denotes the particle position, t is the real time, Δt is the time step, f_i is the temperature distribution function, f_i^{eq} is the equilibrium distribution function, defined as

$$f_i^{\text{eq}} = \omega_i T, \quad i = 0 \sim 4 \quad (9)$$

where ω_i are weight parameters, given by

$$\omega_i = \begin{cases} 1/3, & i = 0 \\ 1/6, & i = 1 \sim 4 \end{cases} \quad (10)$$

In Eq. (8), $\mathbf{e}_i$ is the discrete velocity

$$\mathbf{e}_i = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} c \quad (11)$$

where $c = \Delta x / \Delta t$, in which Δx is the space step. The relation between thermal diffusivity (D) and relaxation time coefficient (τ) is

$$\tau = \frac{D \Delta t}{\varepsilon \Delta x^2} + 0.5 = \frac{\lambda \Delta t}{\varepsilon \rho c_p \Delta x^2} + 0.5 \quad (12)$$

where $\varepsilon = 1/3$ if weight parameters ω_i are defined as in Eq. (10). Then, the local temperature can be obtained by the summation of temperature distribution functions

$$T = \sum f_i \quad (13)$$

For the Robin boundary condition, if the straight boundaries are placed at the middle of two nodes, the unknown incoming distribution functions near the boundaries can be obtained by [31, 34]

$$f_i(\mathbf{x}, t + \Delta t) = \frac{-p \Delta x + q}{p \Delta x + q} \tilde{f}_{\bar{i}}(\mathbf{x}, t) + \frac{2 \omega_i \alpha_3 \Delta x}{p \Delta x + q} \quad (14)$$

where $\tilde{f}$ represents the distribution functions after collision, and subscript $\bar{i}$ denotes the direction opposite to i direction. In Eq. (14), p and q are

$$p = \alpha_1, q = \frac{\varepsilon \alpha_2 \Delta x^2}{D \Delta t} \quad (15)$$

Considering a conjugate heat transfer process in a multicomponent medium, the general boundary condition for the conjugate interface is

$$\alpha_{11}^A T^A + \alpha_{12}^A \frac{\partial T^A}{\partial \mathbf{n}^A} = \alpha_{11}^B T^B + \alpha_{12}^B \frac{\partial T^B}{\partial \mathbf{n}^B} + c_1 \quad (16.a)$$

$$\alpha_{21}^A T^A + \alpha_{22}^A \frac{\partial T^A}{\partial \mathbf{n}^A} = \alpha_{21}^B T^B + \alpha_{22}^B \frac{\partial T^B}{\partial \mathbf{n}^B} + c_2 \quad (16.b)$$

where the subscripts A and B denote component A and component B, respectively. If parameters α_{ij}^A , α_{ij}^B , and c_i are specified, one can recover a particular interfacial boundary conditions. For instance, by taking $\alpha_{11}^A = \alpha_{11}^B = 1$, $\alpha_{12}^A = \alpha_{12}^B = 0$, and $c_1 = 0$ in Eq. (16.a), while $\alpha_{21}^A = \alpha_{21}^B = 0$, $\alpha_{22}^A = -\lambda^A$, $\alpha_{22}^B = \lambda^B$, and $c_2 = 0$ in Eq. (16.b), we have

$$T^A = T^B \quad (17.a)$$

$$-\lambda^A \frac{\partial T^A}{\partial \mathbf{n}^A} = \lambda^B \frac{\partial T^B}{\partial \mathbf{n}^B} \quad (17.b)$$

which means that there is no TCR at the interface. By taking $\alpha_{11}^A = \alpha_{11}^B = 1$, $\alpha_{12}^A = R\lambda^A$, $\alpha_{12}^B = 0$, and $c_1 = 0$ in Eq. (16.a), while $\alpha_{21}^A = \alpha_{21}^B = 0$, $\alpha_{22}^A = -\lambda^A$, $\alpha_{22}^B = \lambda^B$, and $c_2 = 0$ in Eq. (16.b), we have

$$T^A - T^B = R_i \lambda^A \frac{\partial T^A}{\partial \mathbf{n}^A} \quad (18.a)$$

$$-\lambda^A \frac{\partial T^A}{\partial \mathbf{n}^A} = \lambda^B \frac{\partial T^B}{\partial \mathbf{n}^B} \quad (18.b)$$

where R_i denotes the value of interfacial TCR.

Equation (16.a) can be decomposed into two Robin boundary conditions

$$\alpha_{11}^A T^A + \alpha_{12}^A \frac{\partial T^A}{\partial \mathbf{n}^A} = \Phi_1 \quad (19.a)$$

$$\alpha_{11}^B T^B + \alpha_{12}^B \frac{\partial T^B}{\partial \mathbf{n}^B} = \Phi_1 - c_1 \quad (19.b)$$

Thus, the boundary scheme in Eq. (14) can be used for the above two equations [31]

$$f_i(\mathbf{x}^A, t + \Delta t) = c_{11}^A \tilde{f}_i(\mathbf{x}^A, t) + c_{12}^A \Phi_1 \quad (20.a)$$

$$f_i(\mathbf{x}^B, t + \Delta t) = c_{11}^B \tilde{f}_i(\mathbf{x}^B, t) + c_{12}^B (\Phi_1 - c_1) \quad (20.b)$$

with $c_{11}^A = c_{11}^B = \frac{-p\Delta x + q}{p\Delta x + q}$, $c_{12}^A = c_{12}^B = \frac{2\omega_i \Delta x}{p\Delta x + q}$. Combining Eqs. (20.a) and (20.b) yields

$$c_{12}^B f_i(\mathbf{x}^A, t + \Delta t) - c_{12}^A f_i(\mathbf{x}^B, t + \Delta t) = \varphi_1 \quad (21)$$

with

$$\varphi_1 = c_{12}^B c_{11}^A \tilde{f}_i(\mathbf{x}^A, t) - c_{12}^A \left[c_{11}^B \tilde{f}_i(\mathbf{x}^B, t) - c_{12}^B c_1 \right] \quad (22)$$

Likewise, for the boundary condition in Eq. (16.b), we have

$$c_{22}^B f_i(\mathbf{x}^A, t + \Delta t) - c_{22}^A f_i(\mathbf{x}^B, t + \Delta t) = \varphi_2 \quad (23)$$

with

$$\varphi_2 = c_{22}^B c_{21}^A \tilde{f}_i(\mathbf{x}^A, t) - c_{22}^A \left[c_{21}^B \tilde{f}_i(\mathbf{x}^B, t) - c_{22}^B c_2 \right] \quad (24)$$

Combining Eqs. (21) and (23) yields

$$f_i(\mathbf{x}^A, t + \Delta t) = \frac{1}{c_{22}^A c_{12}^B - c_{12}^A c_{22}^B} (c_{22}^A \varphi_1 - c_{12}^A \varphi_2) \quad (25.a)$$

$$f_i(\mathbf{x}^B, t + \Delta t) = \frac{1}{c_{22}^A c_{12}^B - c_{12}^A c_{22}^B} (c_{22}^A \varphi_1 - c_{12}^A \varphi_2) \quad (25.b)$$

3.2. LBM for solid–liquid phase change problem

For heat conduction equation with the solid–liquid phase change, the source term in Eq. (6) can be expressed as

$$S = -\frac{\partial \varphi_l L_f}{\partial t c_p} \quad (26)$$

where L_f is the phase change latent heat, and φ_l is the liquid volume fraction. In the LB modeling, an iterative procedure is necessary for the nonlinear source term [35, 36]

$$f_i^k(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + \Delta t \omega_i \frac{L_f}{c_p} [\varphi_l^{k-1}(\mathbf{x}, t + \Delta t) - \varphi_l(\mathbf{x}, t)] \quad (27)$$

where superscript k represents the k th iteration at each time step. In this work, an enthalpy-based method is adopted to track the solid–liquid interface, and the local enthalpy can be calculated by

$$En^k = c_p T^k + L_f \varphi_l^{k-1} \quad (28)$$

where En^k and T^k are the enthalpy and temperature at k th iteration; φ_l^{k-1} is the liquid fraction of the previous iteration, which can be updated by

$$\varphi_l^k = \begin{cases} 0 & En^k < En_s = c_p T_m \\ \frac{En^k - En_s}{En_l - En_s} & En_s \leq En^k \leq En_l = En_s + L_f \\ 1 & En^k > En_l = En_s + L_f \end{cases} \quad (29)$$

where T_m is the melting temperature; En_s and En_l are enthalpies of the solid and liquid state at the melting temperature, respectively. Such iteration is completed until the temperature and melting fields reach the convergence criterion.

3.3. Boundary conditions

As shown in Figure 1, a Dirichlet condition with $T_H = 1200^\circ\text{C}$ is applied at the top boundary, while a Robin condition is applied at the bottom boundary (back surface):

$$-\lambda \frac{\partial T}{\partial y} \Big|_{y=0} = h(T - T_f) \quad (30)$$

where h is the convective heat transfer coefficient, and T_f is the interior environment temperature. In the x direction, the periodic boundary conditions are applied. Initially, the entire domain is assumed to be 20°C .

4. Validation

4.1. LBM for conjugate heat transfer

Consider a one-dimensional heat transfer process in a dual-layer slab. The two layers have the same thickness 0.5 m and thermal conductivity 1 W/(m·K), while the volumetric specific capacities

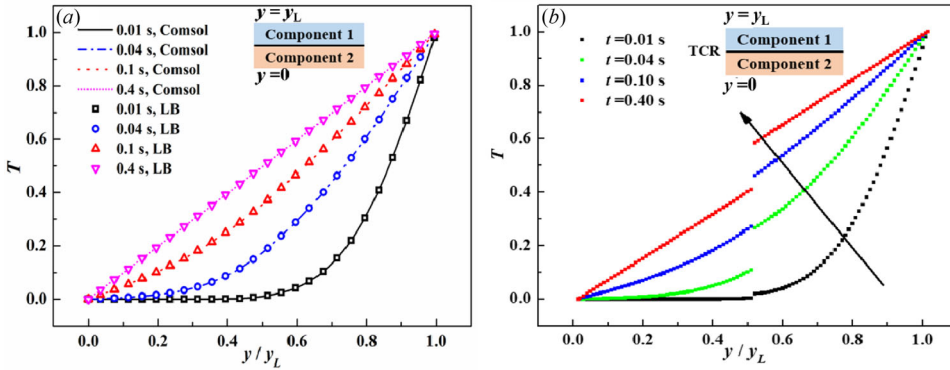


Figure 2. Transient heat transfer in a dual-layer medium: (a) perfect thermal contact; (b) with interfacial TCR = 0.2 m²·K/W.

are 0.5 J/(m³·K) and 1 J/(m³·K), respectively. Initially, the composite slab is at $T=0$, and then, subjected to a high temperature $T=1$ at the top. When the dual layers are in perfect thermal contact, the transient thermal response in the composite slab is shown in Figure 2a. We can see that the temperature profiles obtained between COMSOL 5.5 and present model have a rather good agreement. Conversely, if the interfacial TCR is non-negligible, a significant temperature drop emerges at the interface. When TCR is 0.2 m²·K/W, the transient thermal response in the slab is shown in Figure 2b. At the steady state, the theoretical solution of the interfacial temperature drop can be obtained by

$$\frac{\Delta T}{(T_{\text{up}} - T_{\text{down}})} = \frac{R_i}{R_i + \delta_1/\lambda_1 + \delta_2/\lambda_2} \quad (31)$$

The interfacial temperature drop obtained by the present model is 0.16667 K, which is exactly the theoretical result.

4.2. LBM for solid–liquid phase change problem

In this section, a conduction-induced melting process in a semi-infinite slab is considered. Initially, the entire domain is occupied with the solid phase which is at the melting temperature $T_m=0$. Then, the left side is subjected to a high temperature $T_l=1$, leading to the melting of solid. The position of the melting front (solid–liquid interface) is a function of time [36]

$$x_m(t) = 2\beta\sqrt{Dt} \quad (32)$$

and the temperature field at the fluid domain can be obtained by

$$T(x, t) = T_1 - (T_1 - T_m) \frac{\text{erf}(x/2\sqrt{Dt})}{\text{erf}(\beta)}, \quad 0 \leq x \leq x_m \quad (33)$$

where β is the solution to the transcendental equation

$$\beta \exp(\beta^2) \text{erf}(\beta) = \frac{St}{\sqrt{\pi}} \quad (34)$$

where

$$St = \frac{c_p(T_1 - T_m)}{L_f} \quad (35)$$

In this case, the position of melting front and temperature field obtained by the present model are shown in Figures 3a, b, respectively, which agree well with the theoretical results.

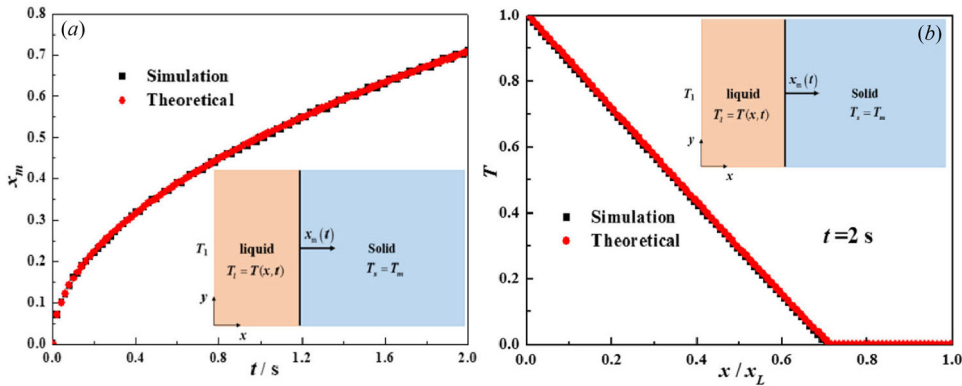


Figure 3. Conduction-induced melting process of a semi-infinite slab: (a) the position of melting front as a function of time and (b) temperature profile in the slab at $t = 2$ s.

5. Results and discussion

5.1. Heat transfer characteristic of the TPS in layout A

5.1.1. Negligible interlayer thermal contact resistance

In this section, the transient heat transfer process of the TPS in the layout A (see Figure 1a) is studied. The total thickness of TPS is 0.05 m. The layer thicknesses of SiO₂ aerogel and that doped with the PCM are denoted as δ_s and δ_p , respectively. To examine effects of the doped PCM on the heat transfer characteristic of the TPS with perfect thermal contact, we compare the thermal response of the TPS without PCM ($\delta_s = 0.035$ m, $\delta_p = 0$) and that of the TPS with PCM ($\delta_s = 0.01$ m, $\delta_p = 0.025$ m). The detailed evolutions of the temperature profile along y -direction of the TPS are shown in Figures 4a, b, respectively, which shows a significant difference in the thermal response, especially in the region of PCM. In the layer of SiO₂ aerogel doped with the PCM, a significant change in the temperature gradient emerges at the melting point 117°C due to the phase change and absorption of latent heat. When the heat transfer approaches the steady state ($t = 12,000$ s), we can observe that the temperature gradient in the layer of the SiO₂ aerogel is the largest due to its extremely low thermal conductivity. Note that the temperature profile is non-linear in the layer of SiO₂ aerogel for the case of Figure 4a since its thermal conductivity is temperature dependent.

The local temperatures at different monitor points ($y = 0, 0.01$ m, 0.02 m, 0.025 m) varying with time are shown in Figure 5. It can be seen that the local temperatures at $y = 0.025$ m, $y = 0.02$ m, and $y = 0.01$ m of the TPS with PCM ($\delta_s = 0.01$ m, $\delta_p = 0.025$ m) are much lower than those of the TPS without the PCM ($\delta_s = 0.035$ m, $\delta_p = 0$) because the melting of PCM would absorb a large amount of energy. Moreover, the temperature differences between two adjacent monitor points are much larger for the TPS without the PCM (i.e., $\Delta T_1 > \Delta T_2$ shown in Figure 5) since the pure SiO₂ aerogel has a smaller thermal conductivity.

Although the doped PCM can absorb heat during the phase transition, it also increases the thermal conductivity of SiO₂ aerogel. To investigate whether the doped PCM can decrease the temperature at the back surface ($y = 0$) or not, we vary the thickness of the layer of SiO₂ aerogel doped with the PCM, δ_p , changing from 0 to 0.035 m with a step of 0.005 m. The temperatures at the back surface of TPSs *versus* time are recorded in Figure 6. It can be seen that the TPS without PCM ($\delta_s = 0.035$ m, $\delta_p = 0$) has the best insulation performance before the moment of $t = 3500$ s, while the TPS with $\delta_p = 0.025$ m ($\delta_s = 0.01$ m) is the optimal thickness design at the moment of $t = 4000$ s. Moreover, the TPS with $\delta_p = 0.015$ m has the worst performance at $t = 2500$ s, while the TPSs with $\delta_p = 0.01$ m and $\delta_p = 0.005$ m are the worst cases for $t = 3500$ s and $t = 4000$ s, respectively. For the TPS in the layout A, the layer of SiO₂ aerogel doped with the

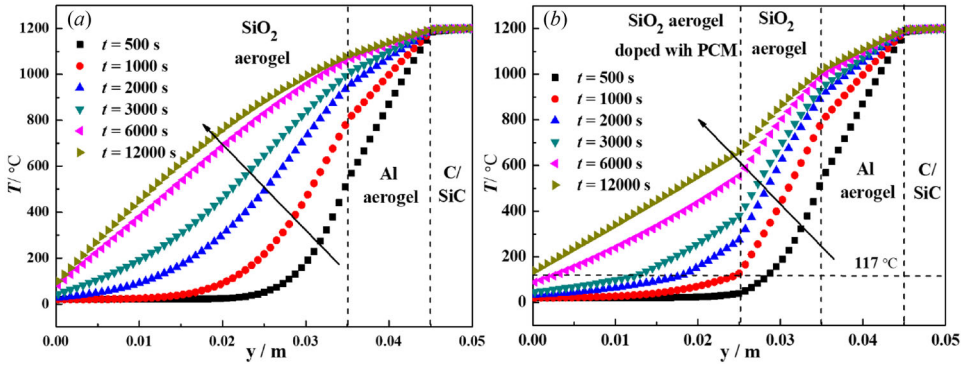


Figure 4. Transient thermal response of the TPS with perfect thermal contact: (a) TPS without PCM ($\delta_s = 0.035$ m, $\delta_p = 0$); (b) TPS with the PCM ($\delta_s = 0.01$ m, $\delta_p = 0.025$ m).

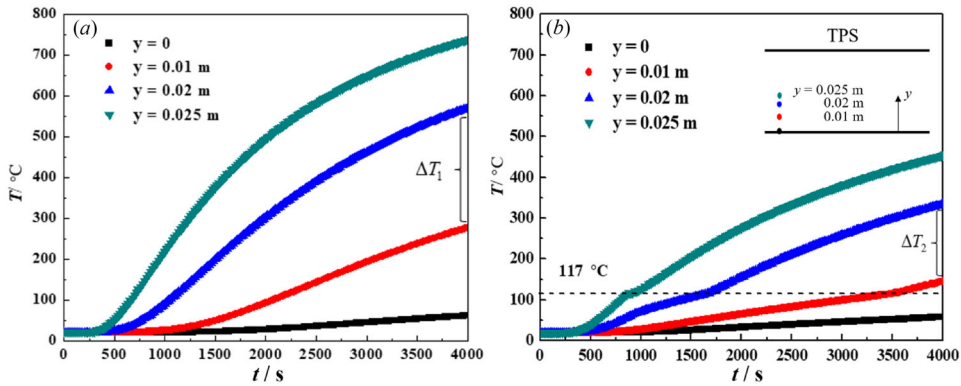


Figure 5. Variations of local temperatures at different monitor points ($y = 0, 0.01$ m, 0.02 m, 0.025 m): (a) TPS without PCM ($\delta_s = 0.035$ m, $\delta_p = 0$); (b) TPS with the PCM ($\delta_s = 0.01$ m, $\delta_p = 0.025$ m).

PCM is at the bottom. Thus, the latent heat of the PCM cannot be fully utilized since the temperature at the back surface of TPS is often required below the melting point. When the benefit obtained by the PCM cannot outperform its adverse effect, the introduction of PCM would deteriorate the thermal performance. Before the moment of $t = 3500$ s, only a small portion of the PCM is melted, which makes the TPS without the PCM have the best thermal performance. As time goes on, more and more PCMs are utilized and their advantages become dominant at the moment of $t = 4000$ s. At different time moments, the detailed melting portions of PCMs in the TPSs with different layer thicknesses δ_p are presented in Table 2. The moment that the PCM in the TPS with $\delta_p = 0.005$ m starts to melt is 4300 s. Therefore, the temperature at the back surface of TPS with $\delta_p = 0.005$ m is always higher than that of the TPS without PCM before $t = 4000$ s (see Figure 6b). When δ_p increases, it takes the advantage of increasing melting portion of the PCM but suffers from the disadvantage of the augmented thermal conductivity. As a result, the temperature at the back surface of TPS shows a non-monotonic trend *versus* the δ_p , as shown in Figure 6b. The thermal performance of the TPS with $\delta_p = 0.035$ m is worse than that of the TPS with $\delta_p = 0.03$ m for the studied moments (i.e., $t = 2500$ s, 3500 s, 4000 s).

5.1.2. Non-negligible interlayer thermal contact resistance

Actually, the TPS consists of multiple layers exists inevitable interlayer TCR. Here, we investigated the transient heat transfer process of the TPS in the layout A with the consideration of the

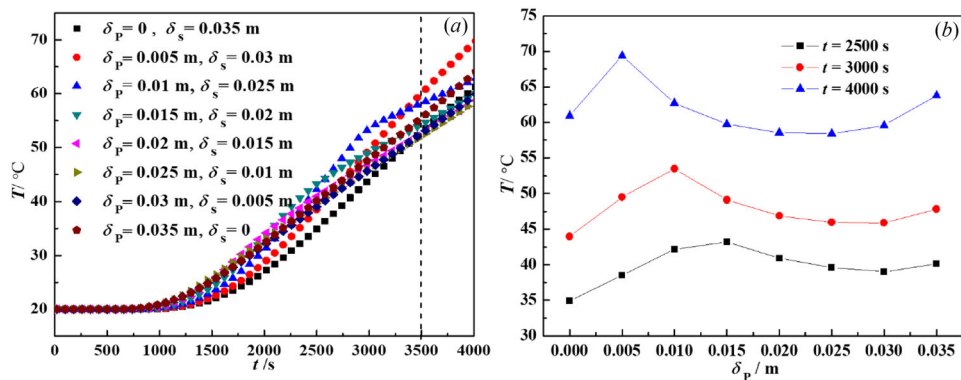


Figure 6. The temperature at the back surface of TPSs in Layout A with different combinations of layer thickness δ_p (SiO₂ aerogel doped with PCM) and δ_s (SiO₂ aerogel): (a) back surface temperatures versus time; (b) back surface temperatures versus δ_p .

Table 2. Melting information of PCM in the TPSs with different layer thicknesses of δ_p (layout A).

	$\delta_p = 0.005\text{ m}$	$\delta_p = 0.01\text{ m}$	$\delta_p = 0.015\text{ m}$	$\delta_p = 0.02\text{ m}$	$\delta_p = 0.025\text{ m}$	$\delta_p = 0.03\text{ m}$	$\delta_p = 0.035\text{ m}$
Start melting moment	4300 s	2900 s	2080 s	1440 s	980 s	520 s	240 s
Melting portion at $t = 2500\text{ s}$	0	0	8.3%	26.25 %	39.0%	50.0%	59.3%
Melting portion at $t = 3000\text{ s}$	0	2.5 %	21.7%	38.75%	50.0%	59.2%	67.9%
Melting portion at $t = 4000\text{ s}$	0	27.5%	46.7%	58.75%	67.0%	74.2%	80.7%

interlayer TCR. In this work, we consider the TCRs between the layers of C/SiC, Al aerogel and SiO₂ aerogel, as shown in Figure 1. The layer thicknesses of TPS are $\delta_c = 0.005\text{ m}$, $\delta_A = 0.01\text{ m}$, $\delta_s = 0.01\text{ m}$, and $\delta_p = 0.025\text{ m}$, respectively. The values of TCR are set to be $R_c = 0.001\text{ (m}^2\cdot\text{K)}/\text{W}$, $R_c = 0.01\text{ (m}^2\cdot\text{K)}/\text{W}$, and $R_c = 0.1\text{ (m}^2\cdot\text{K)}/\text{W}$, to investigate the effects of the TCR on the transient response of the TPS. Figure 7 shows the heat transfer characteristic in the TPS with different values of the TCR. Clearly, the temperature profiles have not much difference between the cases of $R_c = 0.001\text{ (m}^2\cdot\text{K)}/\text{W}$ and $R_c = 0.01\text{ (m}^2\cdot\text{K)}/\text{W}$. However, for the case of $R_c = 0.1\text{ (m}^2\cdot\text{K)}/\text{W}$, an obvious temperature drop exists at the interface, which also decrease the temperature at the back surface of the TPS. Figure 7b shows effects of the interlayer TCR on the variation of temperature at three monitor points inside the TPS. Obviously, when $R_c = 0.1\text{ (m}^2\cdot\text{K)}/\text{W}$, the temperature at the monitor points is much lower than that of the TPS with perfect thermal contact, leading to a significant delay to approach the melting point. For different values of interlayer TCR, the detailed melting information of the PCM in the TPSs are listed in Table 3. We find that the moment that the PCM start to melt has a 640 s delay for the case of $R_c = 0.1\text{ (m}^2\cdot\text{K)}/\text{W}$ while no effects for the case of $R_c = 0.001\text{ (m}^2\cdot\text{K)}/\text{W}$. It is because the interlayer $R_c = 0.001\text{ (m}^2\cdot\text{K)}/\text{W}$ is negligible when compared with the bulk thermal resistance of a multilayer TPS which is evaluated by $\delta_{\text{tot}}/\lambda_{\text{eff}}$ ($R \sim 1\text{ (m}^2\cdot\text{K)}/\text{W}$).

We now examine the thermal performance of the TPS in the layout A when considering the interlayer TCR. By varying the layer thickness of δ_p , the temperature variations at the back surface of TPS with $R_c = 0.1\text{ (m}^2\cdot\text{K)}/\text{W}$ are recorded in Figure 8. Different from the results obtained when TPSs are in perfect thermal contact, the TPS with $\delta_p = 0$ has the best thermal insulation performance even by the moment of $t = 4000\text{ s}$ when $R_c = 0.1\text{ (m}^2\cdot\text{K)}/\text{W}$. It is because the introduced TCR postpones the heat diffusion and reduces the melting portions of the PCM in the TPS. At different time moments, the temperatures at the back surface of TPSs as a function of the layer thickness δ_p are shown in Figure 8b. When $R_c = 0.1\text{ (m}^2\cdot\text{K)}/\text{W}$, the layer thickness δ_p corresponding to the worst thermal performance shifts to a higher value compared to results that the TPS are in perfect thermal contact.

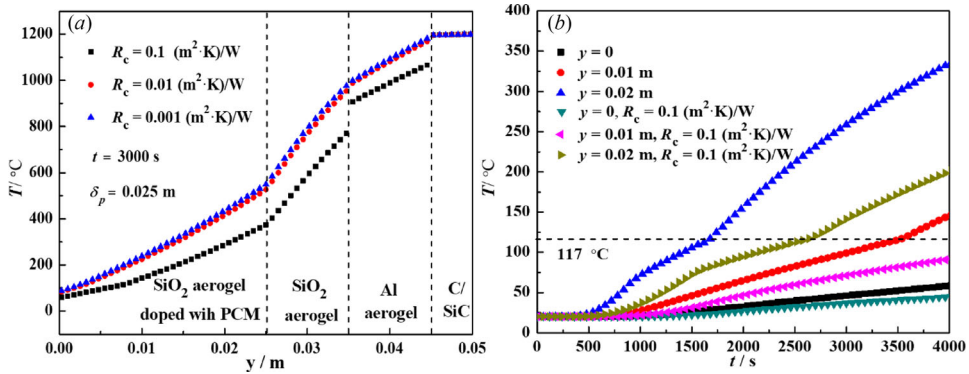


Figure 7. Comparisons of the thermal response of TPSs at different values of the TCR: (a) temperature profile along the y -direction; (b) temperatures at different monitor points.

Table 3. Melting information of PCM in TPSs with different values of interlayer TCR (layout A).

	$R_c = 0$	$R_c = 0.001$ (m ² ·K)/W	$R_c = 0.01$ (m ² ·K)/W	$R_c = 0.1$ (m ² ·K)/W
Start melting moment	980 s	980 s	1040 s	1620 s
Melting portion at $t = 2500$ s	39.0%	39.0%	36.0%	17.0%
Melting portion at $t = 3000$ s	50.0%	49.0%	46.0%	26.0%
Melting portion at $t = 4000$ s	67.0%	67.0%	64.0%	43.0%

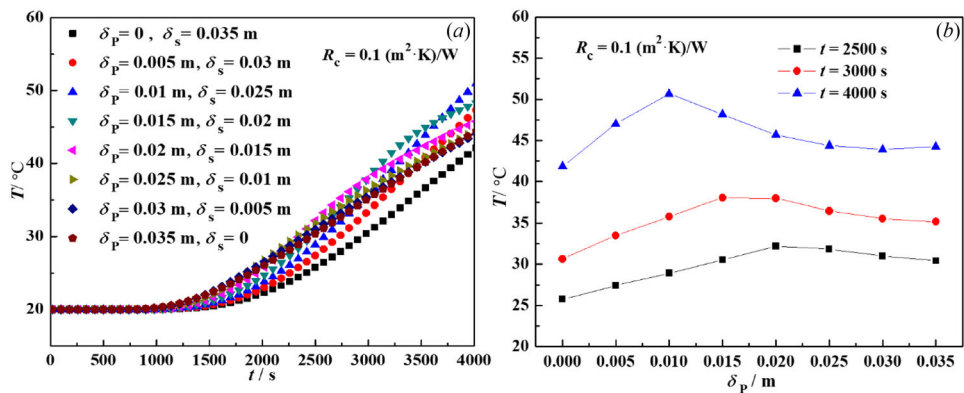


Figure 8. The temperature at the back surface of TPSs in layout A with different combinations of layer thickness δ_p and δ_s when interlayer $R_c = 0.1$ (m²·K)/W: (a) back surface temperatures versus time; (b) back surface temperatures versus δ_p .

5.2. Heat transfer characteristic of the TPS in Layout B

For the TPS in the layout A, the above studies have shown that the doped PCM cannot improve the thermal insulation performance on most occasions. To harness the latent heat of the doped PCM to a great extent, a multilayer TPS in the layout B is designed (the layer of SiO₂ aerogel doped with the PCM is put above the layer of the SiO₂ aerogel), as shown in Figure 1b. Below we studied the thermal performance of the TPS in the layout B.

5.2.1. Negligible interlayer thermal contact resistance

First, we study on the thermal performance of the TPS in perfect thermal contact and put our attention on the temperature at the back surface of the TPS. Figure 9 shows the thermal performances of TPSs with different layer thicknesses δ_p . It can be seen that the TPS with $\delta_p = 0.035$ m has the worst thermal insulation performance before $t = 4000$ s, and the optimal thickness of δ_p

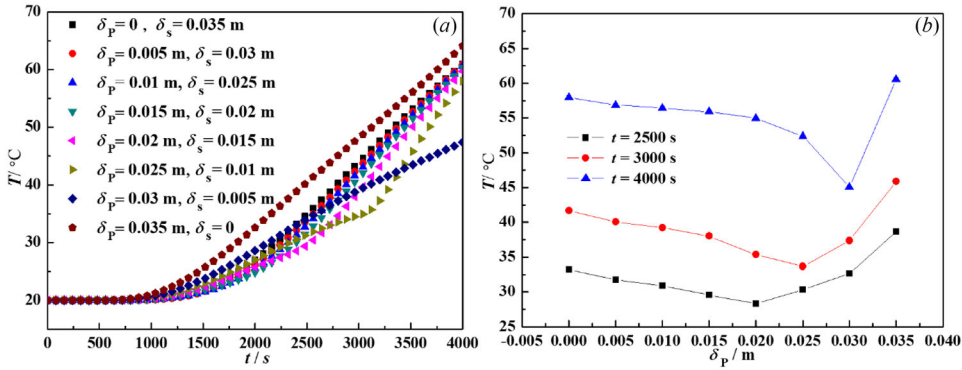


Figure 9. The temperature at the back surface of TPSs in layout B with different combinations of layer thickness δ_p (SiO₂ aerogel doped with PCM) and δ_s (SiO₂ aerogel): (a) back surface temperatures versus time; (b) back surface temperatures versus δ_p .

depends on the service time that the TPS is required to withstand. The TPS with $\delta_p = 0.02$ m has the best thermal insulation performance at $t = 2500$ s, while the TPSs with $\delta_p = 0.025$ m and $\delta_p = 0.03$ m are the best at $t = 3000$ s and $t = 4000$ s, respectively. For a longer service time, a thicker layer of SiO₂ aerogel doped with the PCM is required to provide a sufficient amount of the latent heat. Compared to the TPS in the layout A, the TPS in the layout B has a lower temperature at the back surface because it can utilize a greater proportion of the PCM and take more benefits from the latent heat. The melting information of the TPS with different layer thicknesses δ_p is presented in Table 4. It shows that all the PCM doped in the TPS can be fully utilized before $t = 4000$ s, except the case of the TPS with $\delta_p = 0.035$ m. While the PCM doped in those TPSs with δ_p thinner than the 0.02 m and 0.03 m can be completely utilized before the moment of $t = 2500$ s and $t = 3000$ s, respectively. As a result, the optimal thickness of layer of the SiO₂ aerogel doped with the PCM increases with the service time.

5.2.2. Non-negligible interlayer thermal contact resistance

Here, we study the influences of interlayer TCR on the transient heat transfer process of the TPS in the layout B. Likewise, the values of TCR are set to be $R_c = 0.001$ (m²·K)/W, $R_c = 0.01$ (m²·K)/W, and $R_c = 0.1$ (m²·K)/W, and the layer thicknesses of TPS are $\delta_s = 0.01$ m and $\delta_p = 0.025$ m, respectively. Figure 10 show the thermal performance of TPSs with different values of the TCR. Clearly, when $R_c = 0.1$ (m²·K)/W, there is a significant temperature drop at the interface, and the temperature at the back surface of TPS also decreases. In contrast, the temperature drop at the interface is negligible for the case of $R_c = 0.001$ (m²·K)/W. The altered temperature distribution significantly influences the melting process of the PCM in the TPS, and the detailed melting information of the PCM in TPSs is presented in Table 5. Compared to the results that TPSs are in perfect thermal contact, the TPS with $R_c = 0.001$ (m²·K)/W almost has no influence on the transient heat transfer process, while the TPS with $R_c = 0.1$ (m²·K)/W significantly postpones the heat transfer and the complete melting moment is changed from 2960 s to 4000 s.

Figure 11 shows the temperature at the back surface of the TPS with $R_c = 0.1$ (m²·K)/W at different δ_p . We find that when $R_c = 0.1$ (m²·K)/W, the temperatures at the back surface are decreased compared to the cases without the interlayer TCR. Moreover, the optimal layer thickness δ_p corresponding to the best insulation performance is also reduced (see Figures 11b and 9b). It is because the existence of the TCR postpones the heat diffusion and reduces the melting portions of the PCM, which inevitably affects the optimal δ_p .

Table 4. Melting information of the TPSs with different layer thicknesses of δ_p (Layout B).

	$\delta_p = 0.005$ m	$\delta_p = 0.01$ m	$\delta_p = 0.015$ m	$\delta_p = 0.02$ m	$\delta_p = 0.025$ m	$\delta_p = 0.03$ m	$\delta_p = 0.035$ m
Start melting moment	240 s	240 s	240 s	240 s	240 s	240 s	240 s
Complete melting moment	540 s	960 s	1500 s	2160 s	2960 s	4000 s	6520 s
Melting portion at $t = 2500$ s	100%	100%	100%	100%	86.0%	70.0%	59.3%
Melting portion at $t = 3000$ s	100%	100%	100%	100%	100%	80.8%	67.9%
Melting portion at $t = 4000$ s	100%	100%	100%	100%	100%	100%	80.7%

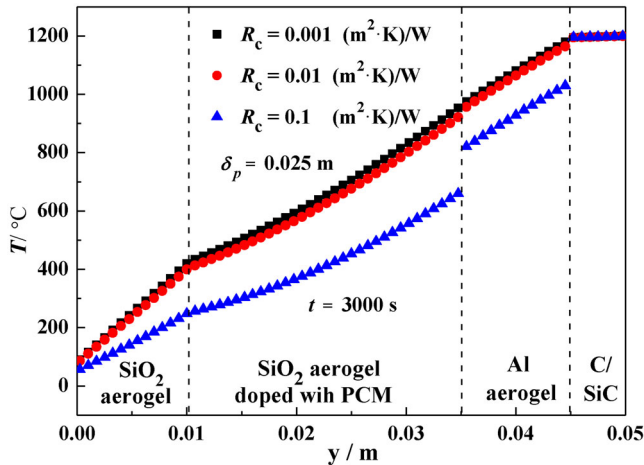


Figure 10. The temperature profile of TPSs with different values of the TCR at $t = 3000$ s.

Table 5. Melting information of PCM in TPSs with different values of interlayer TCR (Layout B).

	$R_c = 0$	$R_c = 0.001$ ($\text{m}^2 \cdot \text{K}/\text{W}$)	$R_c = 0.01$ ($\text{m}^2 \cdot \text{K}/\text{W}$)	$R_c = 0.1$ ($\text{m}^2 \cdot \text{K}/\text{W}$)
Start melting moment	240 s	240 s	280 s	540 s
Complete melting moment	2960 s	2980 s	3140 s	4500 s
Melting portion at $t = 2500$ s	86.0%	85.0%	82.0%	55.0%
Melting portion at $t = 3000$ s	100%	100%	96.0%	68.0%
Melting portion at $t = 4000$ s	100%	100%	100.0%	90.0%

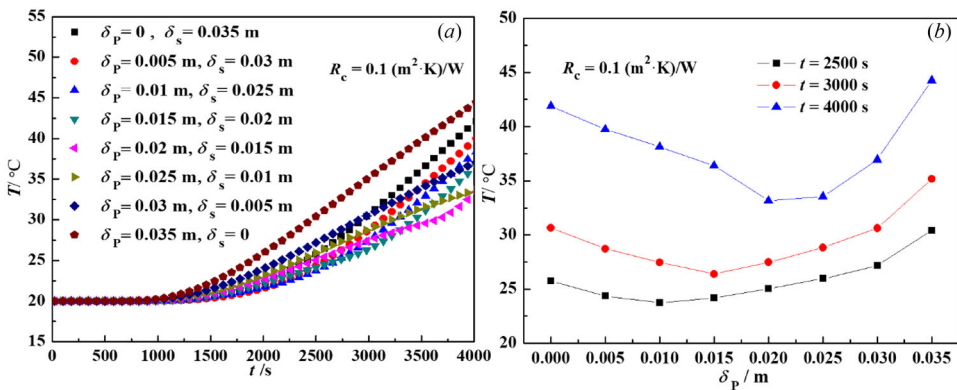


Figure 11. The temperature at the back surface of TPSs in Layout B with different combinations of layer thickness δ_p and δ_s when interlayer $R_c = 0.1(\text{m}^2 \cdot \text{K})/\text{W}$: (a) back surface temperatures versus time; (b) back surface temperatures versus δ_p .

6. Conclusions

In this article, the transient heat transfer characteristics of two kinds of TPSs doped with the phase change materials are analyzed in detail. A thermal lattice Boltzmann model is adopted to consider the effect of interlayer thermal contact resistance on the transient thermal response of the TPS. Meanwhile, an enthalpy-based lattice Boltzmann model is adopted to solve the conduction-induced melting problem. After validations, the effects of the doped PCM and interlayer thermal contact resistance on the transient thermal response of the TPS are investigated. The following conclusions can be drawn

1. The doped PCM significantly influences the transient heat transfer process of the TPS. Whether the doped PCM can improve the thermal insulation performance of the TPS depends on the layout design of the multiple layers and the service time that the TPS is required to withstand;
2. To enhance the thermal performance of the TPS, the layer that SiO₂ aerogel doped with the PCM should be put above the layer of SiO₂ aerogel. There exists an optimal layer thickness of the PCM that makes a TPS has the best thermal insulation performance, and the optimal value increases with the service time;
3. If the interlayer thermal contact resistance is rather small compared to the bulk thermal resistance of the TPS, its effect on the thermal insulation performance of the TPS is negligible; otherwise, the interlayer thermal contact resistance has a significant effect and will reduce the optimal layer thickness that contains the PCM.

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